Mapping the Relationship Between Socioeconomic Vulnerability and Wastewater Discharge in the San Joaquin County

Homework Assignment 1

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```
# Load libraries
library(tidyverse)
library(sf)
library(here)
library(tmap)

# Read in geodatabase of EJScreen data at the Census Block Group level
ejscreen                    <-                    sf::st_read(here::here("data",
"ejscreen","EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"))
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
            `/Users/karlaramos/Desktop/Git/eds223-HW/EDS223-HW1/data/ejscreen/
EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
# Filter to California
california <- ejscreen %>%
  dplyr::filter(ST_ABBREV == "CA")

# Filter to San Joaquin County
san_joaquin <- ejscreen %>%
```

```
  dplyr::filter(CNTY_NAME %in% c("San Joaquin County"))

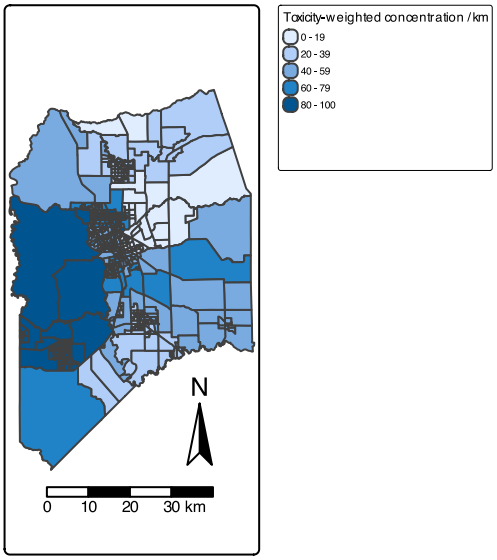
# Find the average values for all variables within counties
california_counties <- aggregate(california, by = list(california$CNTY_NAME),
FUN = mean)
```

```
# Plot percentile of wastewater discharge
sj_waterdischarge <-tm_shape(san_joaquin)+  # Define the spacial object to plot
  tm_title('Waterwater Discharge in San Joaquin County (2023)')+
 tm_polygons(fill= 'P_PWDIS', fill.legend = tm_legend(title = "Toxicity-weighted
concentration / km"))+
  tm_add_legend(type = "line", col = "white") +
  tm_layout(bg.color = "white")+
  tm_style("white") +
  tm_compass() +
  tm_scalebar()
```

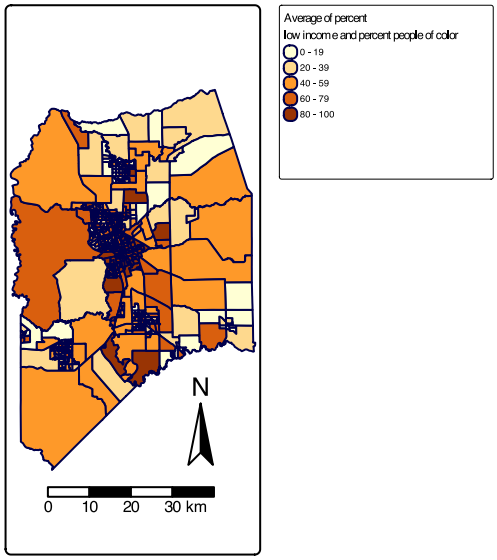
```
# Plot EJScreen Supplemental Demographic Index
# Supplemental Demographic Index is based on the average of two demographic
indicators; Percent Low-Income and Percent Minority.
sj_demographic <- tm_shape(san_joaquin)+ # Define the spacial object to plot
    tm_polygons(fill= 'P_DEMOGIDX_2', fill.legend = tm_legend(title = "Average
of percent
low income and percent people of color")) +
    tm_title('EJ Demographic Index in San Joaquin County (2023)')+
    tm_add_legend(type = "line", col = "white") +
    tm_layout(bg.color = "white")+
    tm_style("beaver") +
    tm_compass(type ="arrow") +
    tm_scalebar() #percentile of wastewater discharge
```

```
# Arrange maps in a grid
tmap_arrange(sj_waterdischarge, sj_demographic)
```

Waterwater Discharge in San Joaquin County (2023)



EJ Demographic Index in San Joaquin County (2023)



Mapping summary:

In San Joaquin County, unsafe water quality poses a major public health threat, particularly for communities identified by EJScreen's demographic indicators. Water contamination and toxicity have been linked to agricultural and industrial activity, but the full scope of exposure is difficult to measure because much of the available data does not account for people who travel through, recreate in, or work near contaminated water.

The map on the left (Wastewater Discharge in San Joaquin County, 2023) shows wastewater discharge from permitted facilities, measured as toxicity-weighted concentration divided by distance in meters. The map on the right (EJ Demographic Index in San Joaquin County, 2023) shows a demographic index based on percent low-income and percent people of color. In the western part of the county, areas with high wastewater toxicity substantially overlap with areas scoring high on the demographic index.